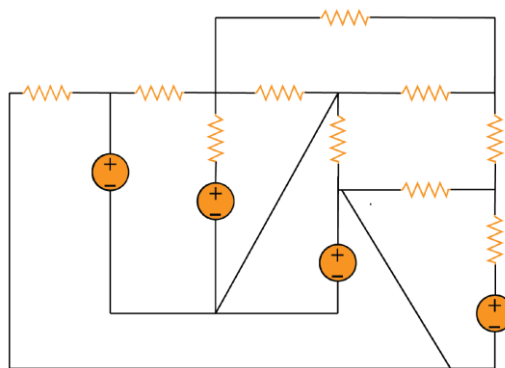


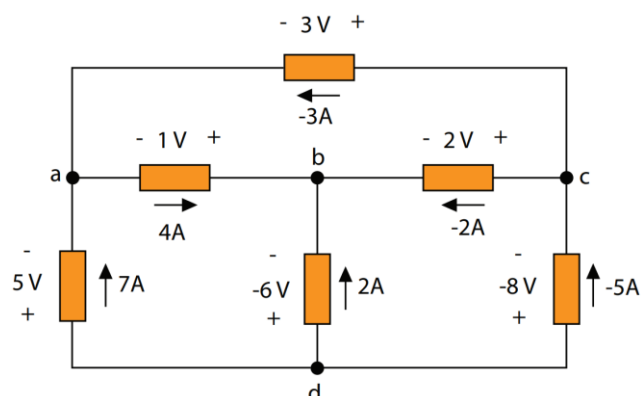
Topic 2: Kirchhoff's Laws, Nodal & Mesh Analysis

1. Identify all the nodes, branches, and meshes in the following figure.



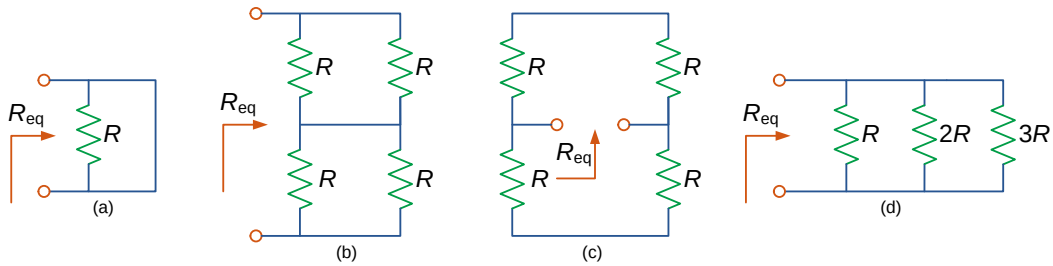
Answer: There are 14 branches, 8 nodes and 7 meshes.

2. In the following circuit (Q2 in Online Tutorial 1) the total sum of powers being consumed and generated is not equal to zero, violating the conservation of energy law.
 - (a) Identify the error in the circuit using Kirchhoff's Current and Voltage Laws.
 - (b) Confirm your answer by calculating the powers and their total sum.



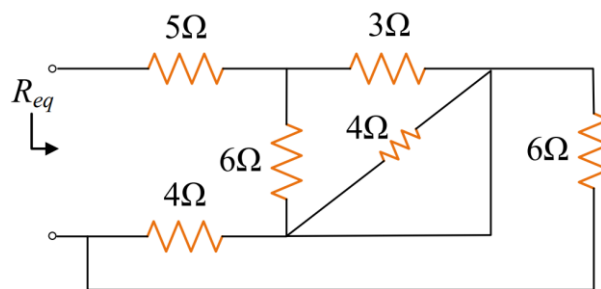
Answer: Branches bd and ad.

3. Find the equivalent resistance of the following networks.



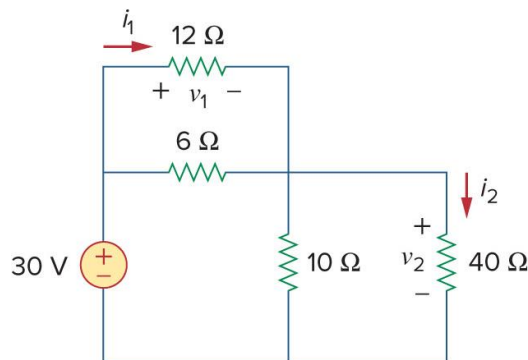
Answer: 0, R , R , $6R/11$

4. (Midterm Exam - S1, 2016) For the circuit below, calculate the equivalent resistance R_{eq} of the network as seen from the terminals



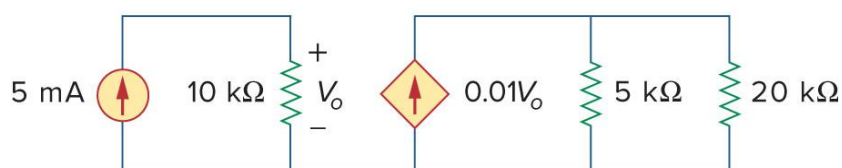
Answer: 9.4Ω

5. Find v_1 and v_2 in the circuit shown below. Also the currents i_1 and i_2 .



Answer: 10 V, 20 V, 833.3 mA, 500 mA

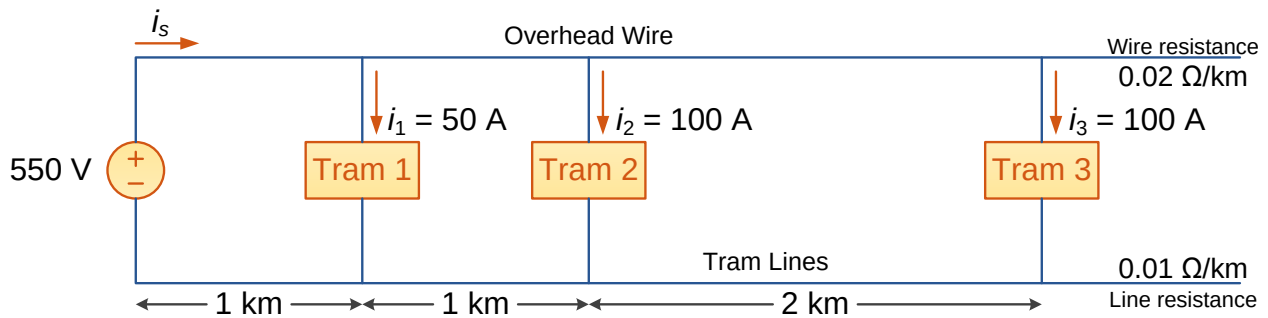
6. In the following circuit, calculate the current and voltage of the $20 \text{ k}\Omega$ resistor. What is the power generated/supplied by the dependent current source?



Answer: 1 kW

7. The DC traction power supply system (TPSS) of a light rail network, as shown in the following figure, consists of a DC Voltage source, overhead wires and an earth return. At a given moment, three trams travel within one section post and absorb 50 A, 100 A, and 100 A, respectively. Considering the overhead wires electric resistance given in units of Ω/km , calculate the power that is absorbed by each trams. What is the overall efficiency of this TPSS?

Efficiency η is defined as the ratio of the total useful power consumed by a system (to perform a task) to the total power supplied to it. Efficiency is an indication of how much of the supplied power is being lost like the heat dissipation by wires resistances.

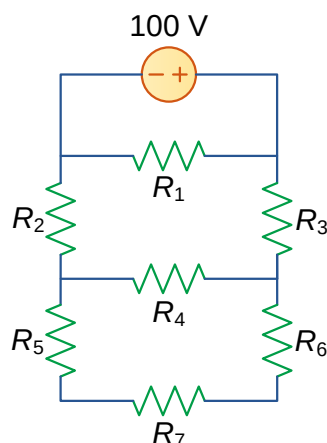


Answer: 97.3%

8. An industrial heating application can be modelled by using a connection of resistive elements as shown in the figure below. The two requirements for this industrial heating application are stated as below,

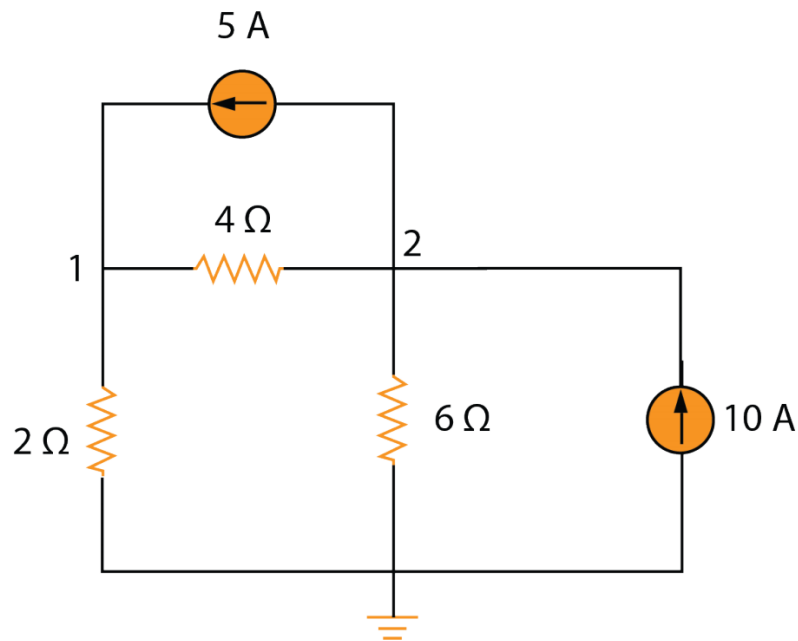
- The total power consumed is equal to 3500 W.
- The elements generate heat uniformly (this means that they dissipate equal power).

Based on the above requirements, calculate the values of the resistances R_1 to R_7 .



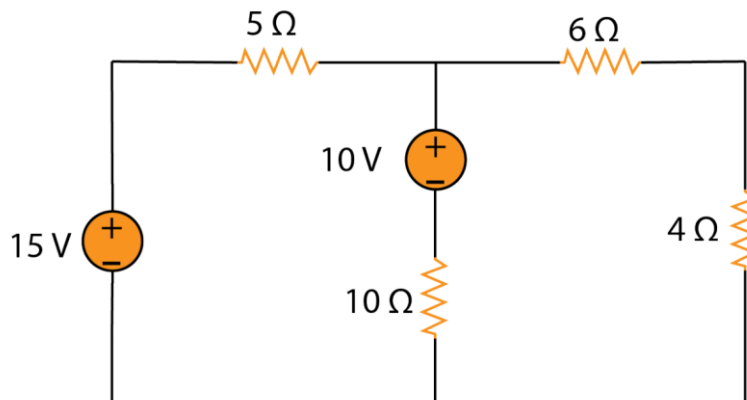
Answer: 20 Ω , 0.56 Ω , 0.56 Ω , 8.89 Ω , 0.987 Ω , 0.987 Ω , 0.987 Ω

9. Find the node voltages in the circuit given below.



Answer: $v_1 = \frac{40}{3} = 13.333 \text{ V}$ and $v_2 = 20 \text{ V}$

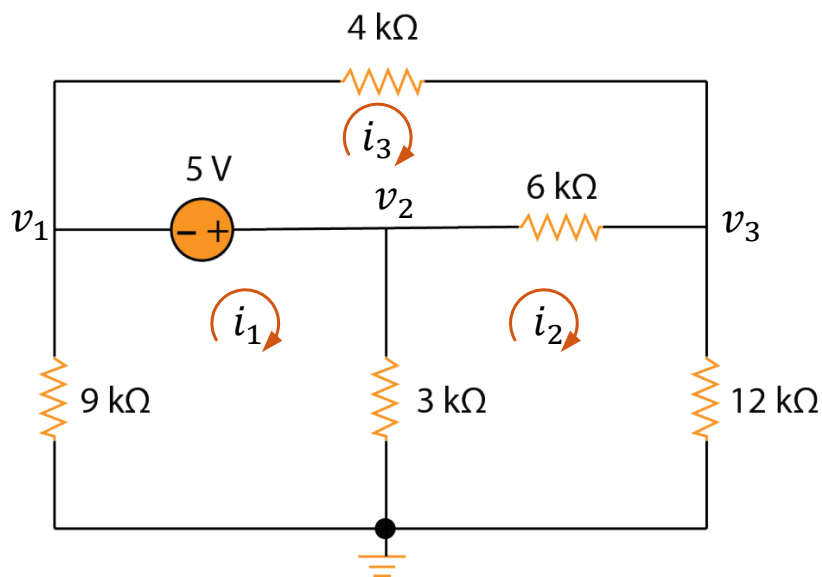
10. In the following circuit, find all the branch currents using mesh analysis. (Assign mesh currents and branch currents individually)



Answer: $I_1 = 1 \text{ A}$ (current in 15-V voltage source and 5-Ω resistor), $I_2 = 1 \text{ A}$ (current in 6-Ω and 4-Ω resistors), and $I_3 = 0 \text{ A}$ (current in 10-V voltage source and 10-Ω resistor).

11. For the circuit below,

- Write nodal equations using nodal analysis.
- Write mesh equations using mesh analysis.



Answer:

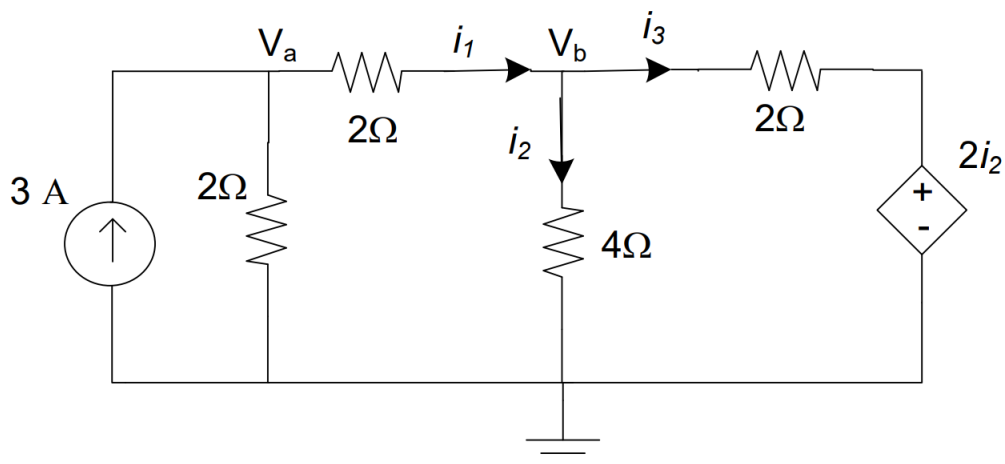
$$\text{a) } \begin{cases} v_1 - v_2 = -5 \\ 13v_1 + 18v_2 - 15v_3 = 0 \\ 3v_1 + 2v_2 - 6v_3 = 0 \end{cases}$$

$$\text{b) } \begin{cases} 12i_1 - 3i_2 = 5 \\ 3i_1 - 21i_2 + 6i_3 = 0 \\ 6i_2 - 10i_3 = 5 \end{cases}$$

12. (Mid-session Exam – S2, 2016)

For the circuit below,

- Apply nodal analysis to write down the node voltage equations at nodes V_a and V_b .
- Solve the voltage equations in part (a) to find the voltages V_a and V_b .
- Find the currents i_1 , i_2 and i_3 based on your results in part (b).



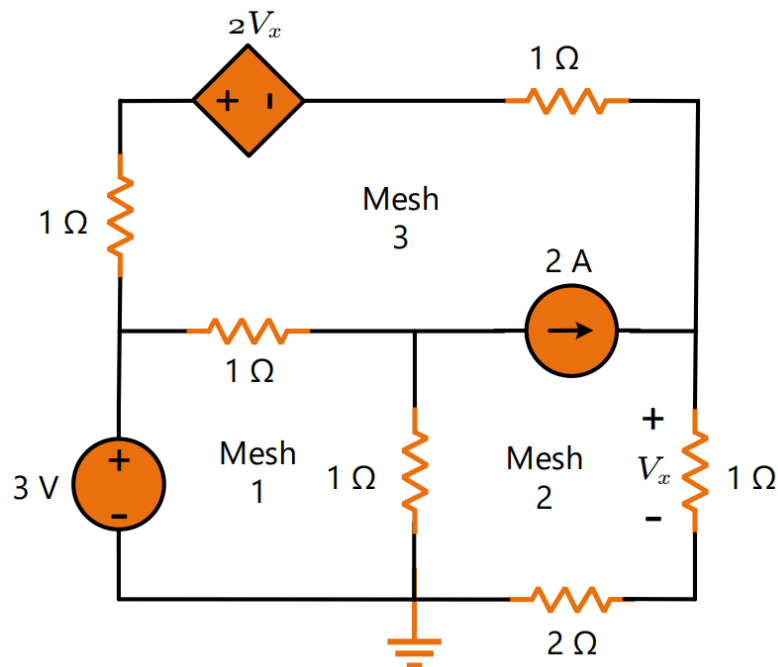
Answer:

- a) $\begin{cases} 2v_a - v_b = 6 \\ 2v_a - 4v_b = 0 \end{cases}$
- b) $V_a = 4 \text{ V}$, and $V_b = 2 \text{ V}$
- c) $i_1 = 1 \text{ A}$, $i_2 = 0.5 \text{ A}$, and $i_3 = 0.5 \text{ A}$

13. (Mid-session Exam – Summer, 2017)

For the circuit below,

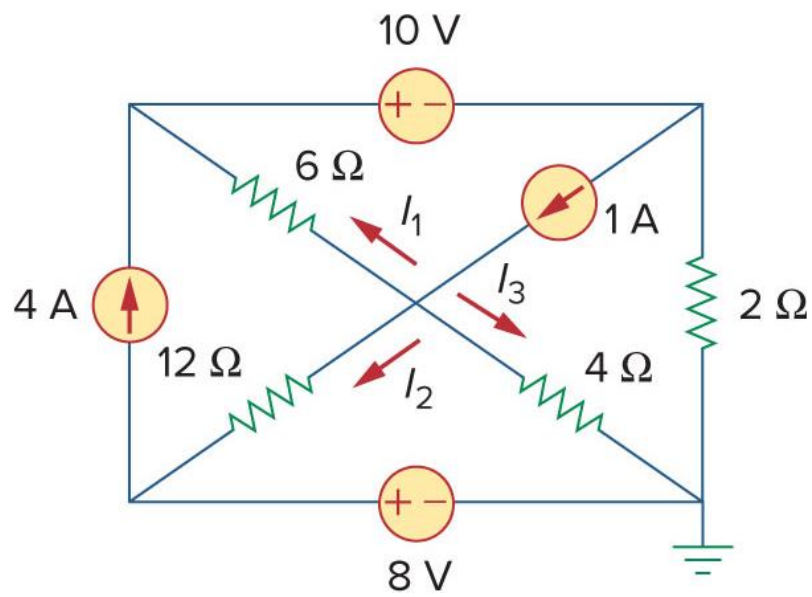
- a) Apply mesh analysis to write down the mesh current equations in Mesh 1, Mesh 2, and Mesh 3 (this part was only given in the actual exam).
- b) Solve the current equations in part (a) to find the mesh current.
- c) Find the power generated by the dependent voltage source based on your results in part (b).



Answer:

- a) $\begin{cases} 2i_1 - i_2 - i_3 = 3 \\ 2i_1 - 6i_2 - 3i_3 = 0 \\ i_2 - i_3 = 2 \end{cases}$
- b) $i_1 = 1.5 \text{ A}$, $i_2 = 1 \text{ A}$, and $i_3 = -1 \text{ A}$
- c) $P = 2 \text{ W}$ supplied

14. In the circuit below, find the branch currents I_1 , I_2 , and I_3 .



Answer: $I_1 = -1$ A, $I_2 = 0$ A, and $I_3 = 2$ A